

SOH Estimation of Li-ion Batteries

Round 2 Progress Report — Steps 1-18

Date: 2026-09-25

Data: NASA BatteryAgingARC-FY08Q4 (B0005, B0006, B0007, B0018), verified raw files tracked in this repository.

Protocol (used identically for every comparison): nested 4-fold LOBO — each battery serves as the unseen outer test in turn; inner validation = first remaining battery; all hyperparameters chosen by inner validation only; SOH in % (Q_{ref} = mean of first 5 discharge cycles); targets standardized on fold-train only; EarlyStopping on inner-val loss; metrics MAE/RMSE/ R^2 + nRMSE in %SOH; seeds {42, 7, 123}.

Leakage controls (every run): structural assert (scaler statistics equal fold-train statistics) + feature-perturbation probes (outer test $\times 2 \rightarrow$ retrain $\rightarrow$ train metrics must not move; three probes across batches, all PASS with $\text{diff} \leq 3.8\text{e-}05$, one exactly 0).

1. Status of the 18-Step Sequence

Step	Task (advisor Part)	Result	Evidence		
2	Verify NASA EIS variables and frequency ordering (A1-A2, M-A/B)	CSV = Rectified_Impedance, exact on all 34,593 rows (max diff < 1e-15 Ohm); row order consistent with low→high frequency (887/887 spectra, indirect evidence); no per-point frequency exists in the distributed .mat — any frequency column is a declared assumption. The flat Nyquist arc is a property of NASA's Rectified_Impedance itself (Case 2): max \	Im\	$\approx 2\text{--}6\text{ mOhm}$ across all 887 spectra, while the raw Battery impedance differs by 32–68× and contains values that are not physically plausible	eis_verify/EIS_Raw_Verification.md
4	Frequency-ordered table + plots; spectrum integrity (A3-A5, M-C/D)	39 points $\times$ 887 spectra = 34,593 CSV rows; timestamps monotonic; 0 NaN/Inf/duplicates; no isolated outlier points	same + A3_*, 3plots.png, A7_*		
	Re-evaluate interpolation with a quantitative measure (A8)	Chord-envelope overshoot: PCHIP 0 in all 1,774 checks; Cubic 0.25–1.76 mOhm mean, worst 11.2 mOhm (exceeds the full reactive span 2–6 mOhm); Linear also 0 but not smooth	Interpolation_Reeval.md		
	Expanded AE reconstruction metrics (D3)	Re(Z) RMSE / Im(Z) RMSE separately + nRMSE + relative error, per battery (table in Section 4); reproduces the B0006 Re RMSE anomaly (14.6 mOhm, $\sim 13\times$ the other batteries)	ae_reconstruction_d3_metrics.csv		
	Recheck Figure 6 (H2)	AE reconstructs the interpolated input , not the measurement (recon-vs-measured = recon-vs-input + $\sim 0.1\text{ mOhm}$); B0006 anomaly is AE-specific	Step6_Fig6_recheck.png		
	Nested 4-fold LOBO (C1-C3)	Implemented; inner validation selects different configs per fold; leakage probes PASS	Steps_7_8_Nested_LOBO.md		
	Tolerance 1/2/3 on downstream metrics (E1-E3)	Tolerances indistinguishable downstream (RMSE 8.57–8.89 %SOH, all within 1 std) $\rightarrow$ pre-declared rule selects tolerance 1 ; gap distribution reported (bimodal: gap 1 or 3 for B0005–07)	same		
	Raw EIS / PCA / AE $\rightarrow$ BPNN (F1.3–4)	Working folds: PCA best (RMSE 5.92, $R^2 +0.61$) ; AE loses to PCA on all three	Steps_9_10_Baselines.md		
	Mean + Linear baselines (F1.1–2)	Mean predictor marks the collapse baseline ($R^2 -0.77$); 256-dim Ridge is fragile (disclosed breaches)	same		
	MAE/RMSE/ R^2 for every unseen battery (D2)	Per-fold tables in both reports above	—		

ep	Task (advisor Part)	Result	Evidence		
	Explain Figure 10 quantitatively (G1-G3)	Old RMSE = $1.126 \times \text{std}(y) \rightarrow$ statistically a flat line, 12.6% worse than the trivial mean; cause chain: unstable latents (correlation flip) + unstandardized target + fixed epochs + single split	Steps_12_Fig10_Collapse.md		
	mean $\pm$ std across folds and seeds (C3)	All aggregates use 12 runs (4 folds $\times$ 3 seeds)	—		
-15	Time-domain baseline and comparison (B1-B5)	TD-BPNN: RMSE 3.59, R^2 +0.85, positive on ALL four folds including B0018 (+0.79) ; EIS pipelines negative on B0018 (-1.0 to -1.2)	Steps_14_16_TimeDomain.md		
	Justify EIS vs time-domain (B6)	Measured pattern matches the B6 criterion verbatim: $R^2_{\text{time}} \gg 0$ while $R^2_{\text{EIS}} < 0 \rightarrow$ the cross-battery failure is an EIS representation problem, not a model problem ; the BPNN architecture is cleared	same		
	EarlyStopping + ModelCheckpoint (I1)	EarlyStopping verified (AE stops at 129–192 epochs); ModelCheckpoint demonstrated: best weights saved at epoch 325/355, reloaded, metrics reproduce exactly (test RMSE 2.5210, R^2 0.9402)	checkpoints/step17_checkpoint.py		
	Reconstruct all results with the corrected pipeline	This report + the regenerated figures below	figures/		

2. Headline result (Step 18 figure)

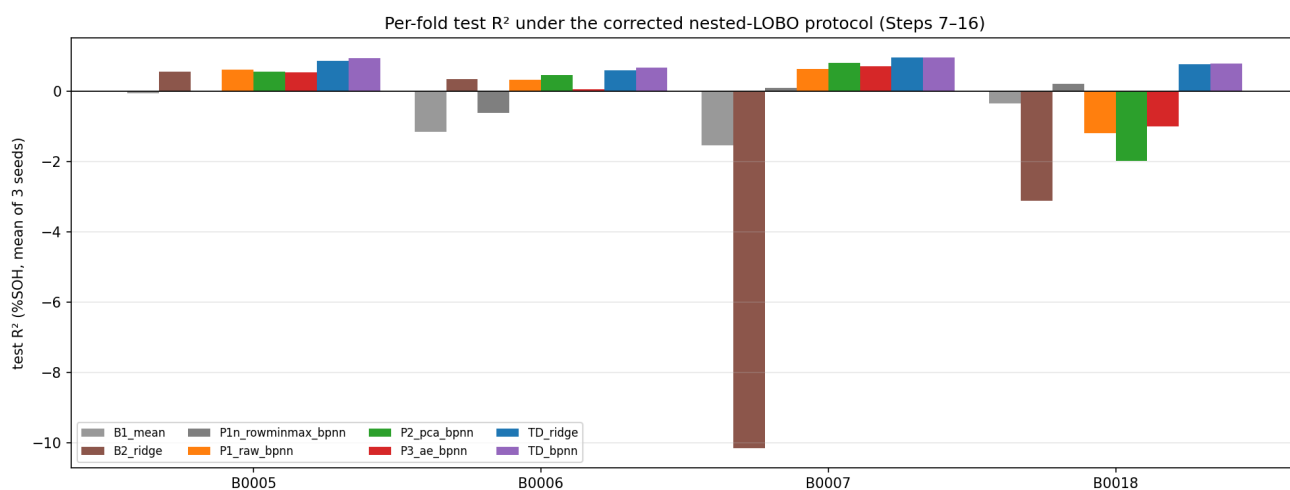


Figure R1. Per-fold test R^2 under the corrected nested-LOBO protocol (mean of 3 seeds). The old collapse ($R^2 \approx 0$ or negative everywhere) is removed on three folds by the corrected EIS protocol, and on all four folds by time-domain features. Reference: raw measured data (Steps 1–4); time-domain features carry the exact same-cycle capacity label.

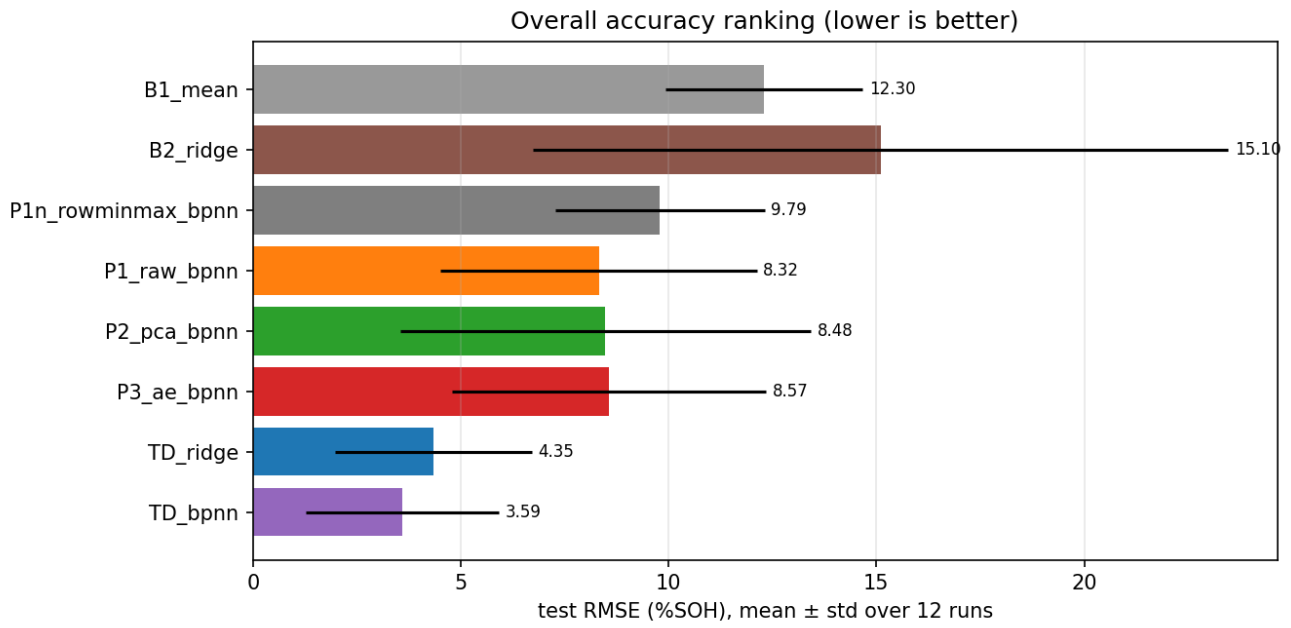


Figure R2. Overall test RMSE (%SOH, mean $\pm$ std over 12 runs). Time-domain features halve the best EIS error.

2b. Steps 1-6 evidence (figures reconstructed from verified raw data)

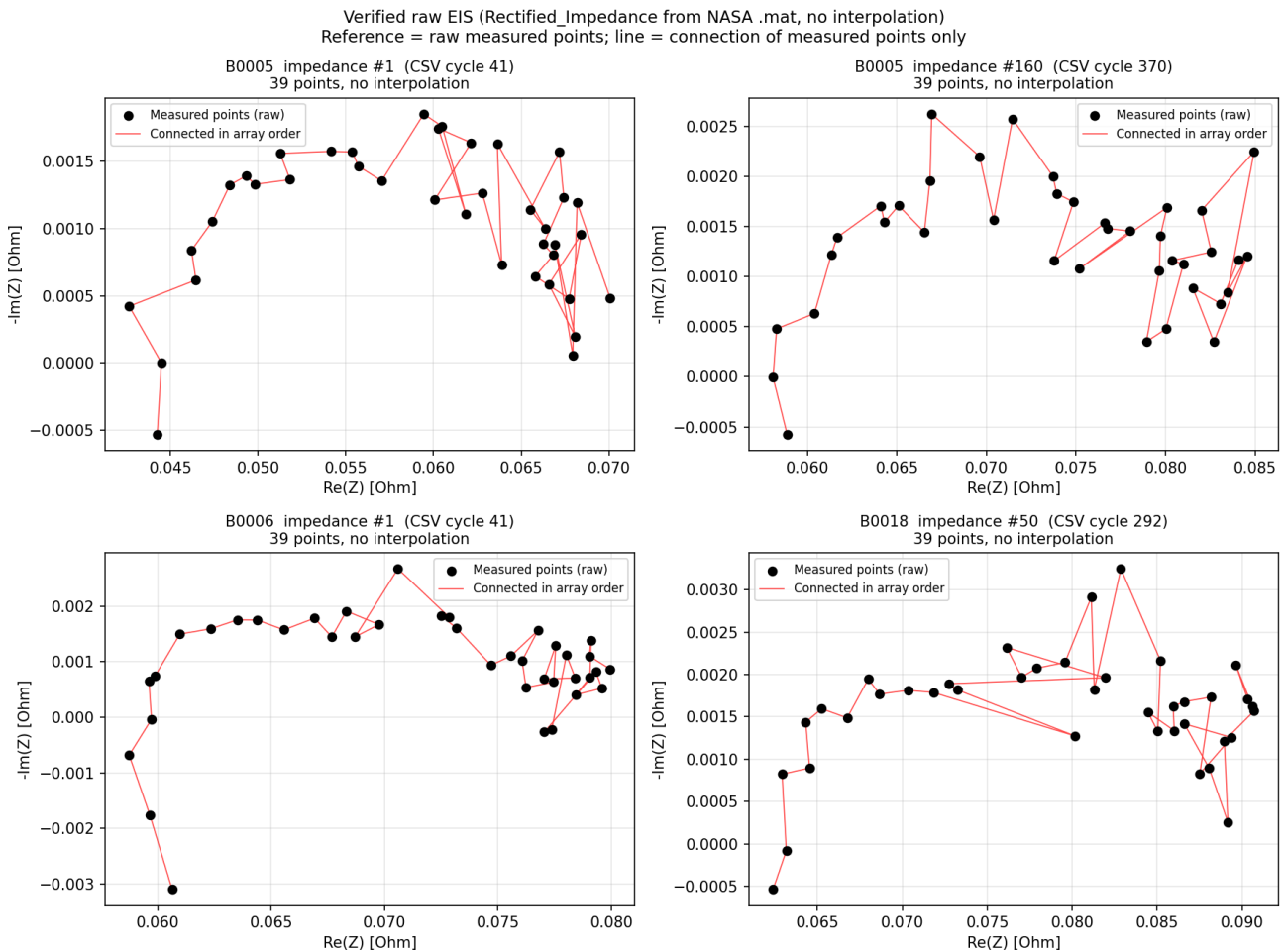


Figure F1. Raw measured EIS points (no interpolation) for 4 spectra across 3 batteries, connected in the verified frequency order (Steps 3-4, A7).

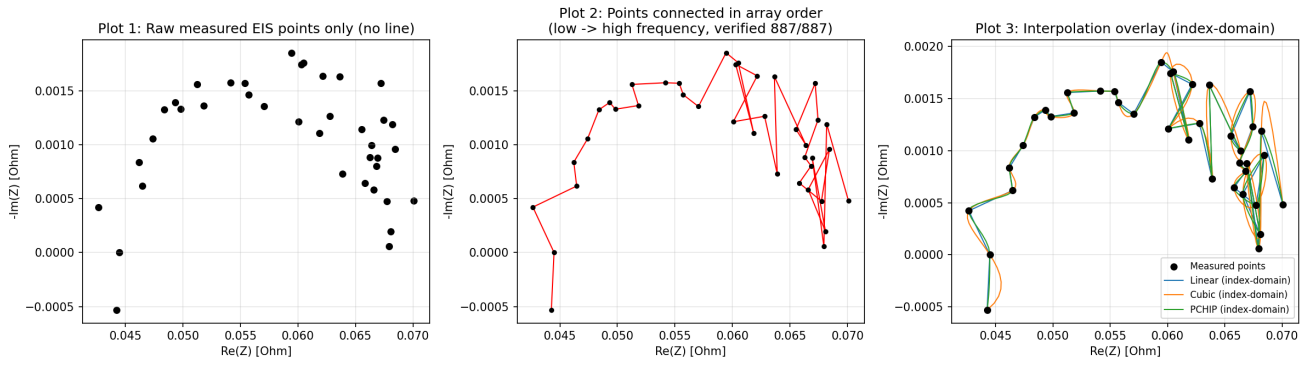


Figure F2. B0005 cycle 41: raw points only / points connected in array order / interpolation overlay (Steps 3-4, A3, M-C).

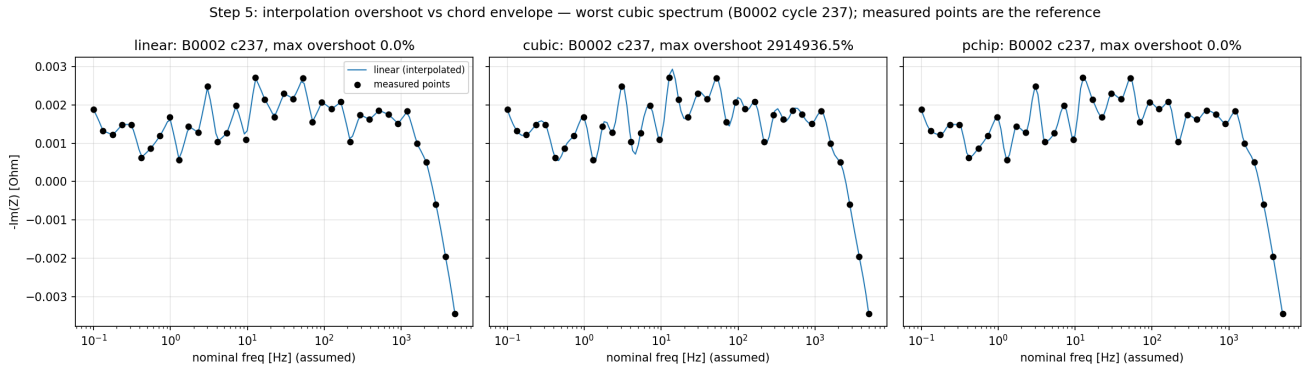


Figure F3. Worst cubic-spline spectrum vs Linear/Cubic/PCHIP (Step 5, A8): cubic overshoots beyond the full reactive signal span; PCHIP stays inside the chord envelope.

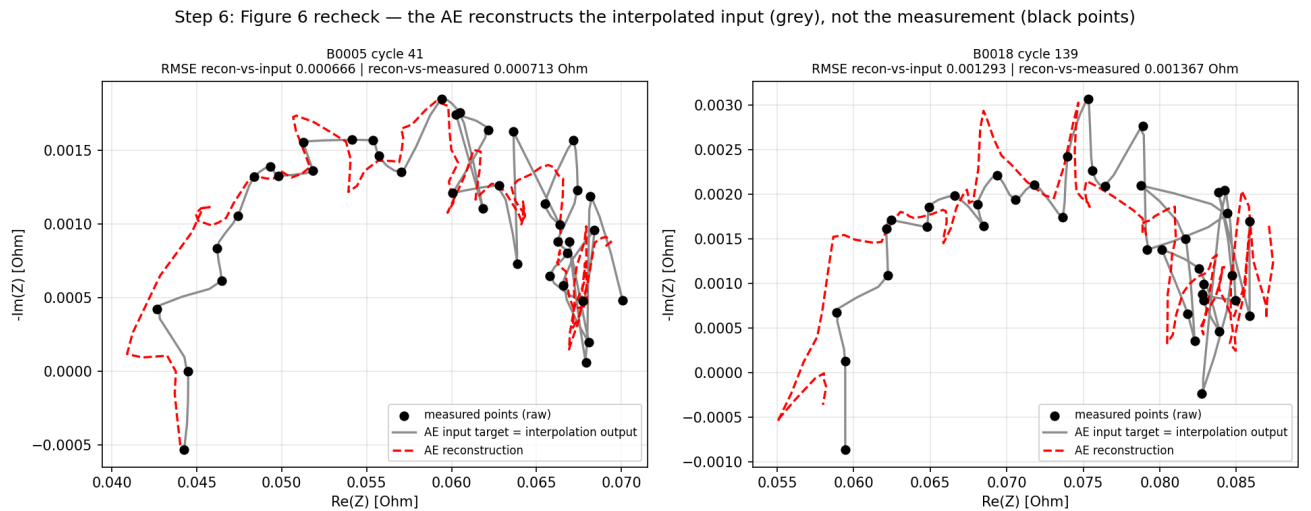


Figure F4. Figure 6 recheck (Step 6, H2): the AE reconstructs the interpolated input target (grey), not the measurement (black markers).

Expanded AE reconstruction metrics (Part D3)

Battery	Re(Z) RMSE ($\pm$ std, Ohm)	Im(Z) RMSE ($\pm$ std, Ohm)	nRMSE	Relative error % ($\pm$ std)
Train B0005	0.00111 $\pm$ 0.00028	0.00023 $\pm$ 0.00007	0.064	22.2 $\pm$ 24.7
Train B0006	0.01461 $\pm$ 0.00234	0.00090 $\pm$ 0.00016	0.519	58.2 $\pm$ 64.4
Val B0007	0.00163 $\pm$ 0.00087	0.00027 $\pm$ 0.00009	0.075	27.7 $\pm$ 41.0
Test B0018	0.00129 $\pm$ 0.00030	0.00034 $\pm$ 0.00009	0.144	23.3 $\pm$ 20.7

The B0006 anomaly appears in Re(Z) RMSE ($\approx 13\times$ the other batteries) with identical scaling — consistent with the AE-specific latent degradation reported in Steps 6 and 9-10. High relative error percentages reflect the tiny absolute scale of this rectified arc (denominators of 1-6 mOhm), not large absolute errors.

2c. Ablations (Part F3 / I3) — completed

All previously pending ablations are now executed under the corrected nested protocol (full tables in [Steps_F3_I3_B7_Ablations.md](#)):

Ablation	Result
AE bottleneck sizes {3, 7, 17, 37, 67}	RMSE 8.30-8.82 %SOH, all within 1 std — indistinguishable; 17 retained
BPNN loss MSE/MAE/Huber (I3)	RMSE 8.32-8.57, indistinguishable — no loss changes the conclusion
Interpolation 6 grid×method configs	RMSE 7.99-8.62, indistinguishable — cubic's Step-5 overshoot has no measured downstream penalty; PCHIP kept for shape preservation

B7 supporting evidence (added): Pipeline E fusion (time-domain + causal backward-aligned EIS latent) reaches R^2 **+0.726**, positive on **all four folds including B0018 (+0.67)**, and beats every EIS-only pipeline — the EIS latent adds value only when anchored by time-domain features. AE applied to time-domain features also does not help (R^2 +0.43 vs +0.85 for raw TD features). The complete pipeline ranking across 10 tested configurations: **TD-BPNN (3.59) > TD-Ridge (4.35) > E-fusion (4.89) > EIS pipelines (8.3-8.6) > Mean (11.5)**.

3. Findings for the advisor's decision (B7)

The measured evidence orders the three information sources cross-battery: **time-domain > EIS(PCA) > EIS(raw) $\approx$ EIS(AE)**. Three constructive directions remain open, per the advisor's B5/B7 framing — the choice is an advisor-level decision:

1. **Pipeline E fusion** (time-domain + EIS features in one model; Part B5 optional row).
2. **AE applied to time-domain features** (keeps the thesis architecture "Two-Stage AE + BPNN", changes the input modality; the current AE-on-EIS result becomes the ablation evidence that motivated it).
3. **EIS as a secondary/complementary modality** with the amplitude domain shift documented (Steps 9-10: per-row normalization rescues B0018 but costs the other folds).

Supporting evidence collected for that discussion: the B0006 AE-specific degradation (Step 6/9-10), the B0018 amplitude domain shift (Step 9-10 P1n), and the B6 verdict above.

4. Declared limitations

- Per-point EIS frequencies are not recoverable from the distributed `.mat`; the interpolation grid rests on the documented 0.1 Hz-5 kHz sweep (declared assumption, Step 1-2).
- Time-domain features describe a completed discharge cycle — practical for cycle-level SOH logging (the BMS records every V/I/T cycle), not a mid-discharge estimator; the full BMS practicality comparison (Part J) is delivered in [Part_J_BMS_Practicality.md](#).
- ICA/DVA and pulse-resistance features were not built (granularity insufficient) — declared, not skipped.
- SOH can exceed 100% by up to 0.8% (measurement variability vs the early-life reference); B0006 fades to 57% (NASA ran past EOL) — data kept as measured.
- Gate breaches in individual runs were disclosed in each step report (Steps 7-10) and none of the working-fold conclusions depend on them.

5. Artifacts

Folder	Content
Autoencoder/eis_verify/	Steps 1-6: verification scripts, raw Nyquist figures, verified dataset, EIS_Raw_Verification.md , Interpolation_Reeval.md
Autoencoder/nested_lobo/	Steps 7-17: mapped datasets, harnesses, all results CSVs, gates, probes, checkpoint, step reports, figures/ (this report's images)

Folder	Content
NASA DataSet/1. BatteryAgingARC-FY08Q4/	Verified source data (tracked in Git since Steps 1–4)